Statistics 280 — Practice Midterm 1A

Unofficial practice exam, written in the style of the W25 and W26 Midterm 1.
Covers Part 1: study design, descriptive statistics, the normal curve, Markov & Chebyshev, correlation, regression.

Name _____ Start time / End time _____

- Answer all questions in the space provided and indicate where you have written additional work.
- Solutions without justification will be marked as wrong for all questions except the scatter-plot matching question. Show all work. A calculator command does not count as justification.
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution. Closed book, no notes. A calculator that can take square roots is allowed.
- Suggested time: 80 minutes. Total: 200 points.

Question	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
Points	30	30	35	35	30	40	200
Score							

1. (30 points) A city's most popular charter middle school has more applicants than seats, so by law it admits students by lottery. One year, 2,000 fifth-graders applied for 600 seats, and the 600 winners were drawn at random from the applicants. Another 8,000 fifth-graders in the district did not apply. Three years later, researchers compared scores on the 8th-grade state math exam, in standard units relative to the whole district:

Group	Number of students	Average score (standard units)
Lottery winners	600	0.35
Lottery losers	1,400	0.20
Did not apply	8,000	−0.10

a) (8 points) A newspaper compares the lottery winners to the students who did not apply and reports that attending the charter school raises math scores by 0.45 standard units. Explain why this comparison is biased. Is the bias likely to be in favor of or against the charter school? Use the numbers in the table.

b) (8 points) Which comparison gives a better estimate of the effect of winning a seat? Give the estimate. Is this comparison a randomized controlled experiment, a natural experiment, or an observational study? Explain briefly.

c) (7 points) Of the 600 winners, 150 declined their seats and attended their neighborhood schools. A researcher drops these 150 students and compares the 450 winners who actually attended to the 1,400 lottery losers. Explain why this comparison could be biased even though the lottery was random. Which study from lecture or the textbook had the same issue?

d) (7 points) The school board asks whether expanding the charter school to all 10,000 students would raise district scores by about the amount you estimated in part b). Give one reason to be cautious.

2. (30 points) Two hospitals perform the same surgery. Patients arrive in either good or poor condition.

	Good condition		Poor condition	
Hospital	**Patients**	**Survived**	**Patients**	**Survived**
A	200	196	800	640
B	800	776	200	150

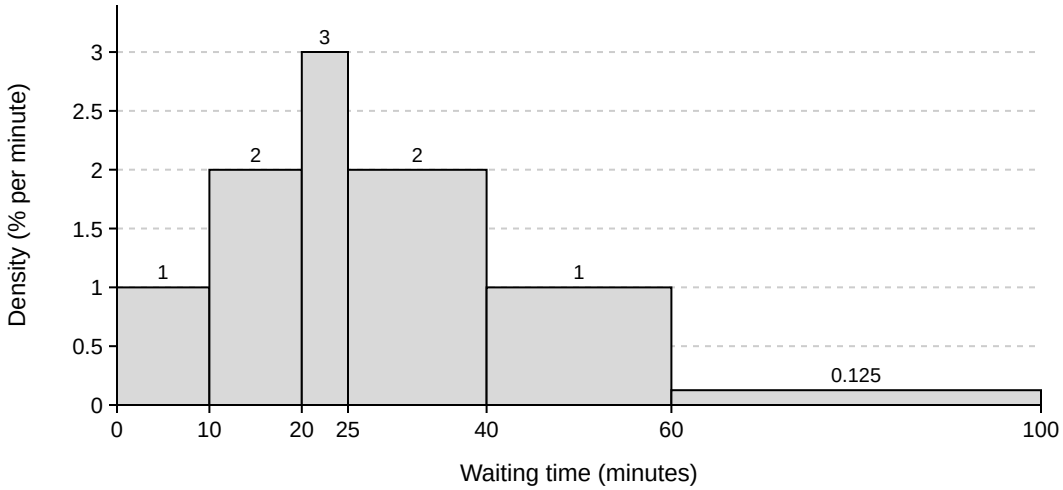
a) (6 points) Find the survival rate for each condition at each hospital, and the overall survival rate at each hospital.

b) (7 points) Which hospital would you choose for this surgery? Explain the apparent contradiction in part a). Then compute each hospital's survival rate for a standard population of patients that is 50% good condition and 50% poor condition.

c) (10 points) A list of 8 *whole numbers* has minimum 2, 25th percentile 5, median 7, 75th percentile 11, and maximum 15, using the definition of percentile from class. The number 11 appears more often than any other number in the list (it is the unique mode), and the mean of the list is 8.5. Find the list, and explain why it is the only possibility.

d) (7 points) All numbers in a list are between 0 and 10. What is the largest possible standard deviation of the list? Give a list that attains it and prove that no list can do better. (Hint: for $0 \leq x \leq 10$, $x^2 \leq 10x$. Use the shortcut formula for the variance.)

3. (35 points) The histogram below shows the distribution of waiting times at a clinic. The number above each block is its height in percent per minute. Assume that values are evenly distributed within each block.



a) (5 points) What percentage of waiting times are between 15 and 30 minutes?

b) (8 points) Estimate the median and the mean. Which is larger, and what feature of the histogram explains this?

c) (7 points) Estimate the interquartile range and the 90th percentile.

d) (7 points) Find the value of c that minimizes (i) $\sum_i |x_i - 2c|$ and (ii) $\sum_i (x_i - c - 10)^2$, where the sums run over all the waiting times x_i in the list.

e) (8 points) The clinic converts each waiting time x into a satisfaction score $y = 100 - 2x$. Find the median, mean, and IQR of the satisfaction scores. How does the SD of the satisfaction scores compare to the SD of the waiting times?

4. (35 points) Commute times for employees at a large company average 30 minutes with a standard deviation of 8 minutes. For parts a) and b), assume that commute times follow the normal curve.

a) (5 points) Estimate the percentage of employees whose commutes are between 20 and 44 minutes.

b) (5 points) Estimate the interquartile range of commute times.

c) (7 points) In a different city, 10% of commutes are longer than 45 minutes and 20% are shorter than 20 minutes. Assuming commute times there follow the normal curve, estimate their mean and standard deviation.

For parts d) – f), scores on an exam are between 0 and 100, with mean 72 and SD 12. Do not assume that the scores follow the normal curve.

d) (6 points) Bound the percentage of students scoring 90 or above, first with Markov's inequality and then with Chebyshev's inequality. Which bound is more useful here, and why?

e) (7 points) (i) Using only the mean and the fact that no score exceeds 100 (not the SD), find the largest possible percentage of students scoring 40 or below, and describe a list of scores that attains it. (Hint: apply Markov's inequality to 100 − score.) (ii) Now use the SD as well. What bound do you get?

f) (5 points) Find the largest possible percentage of students scoring below 48 or above 99.

5. (30 points)

a) (8 points) Find the correlation between x and y:

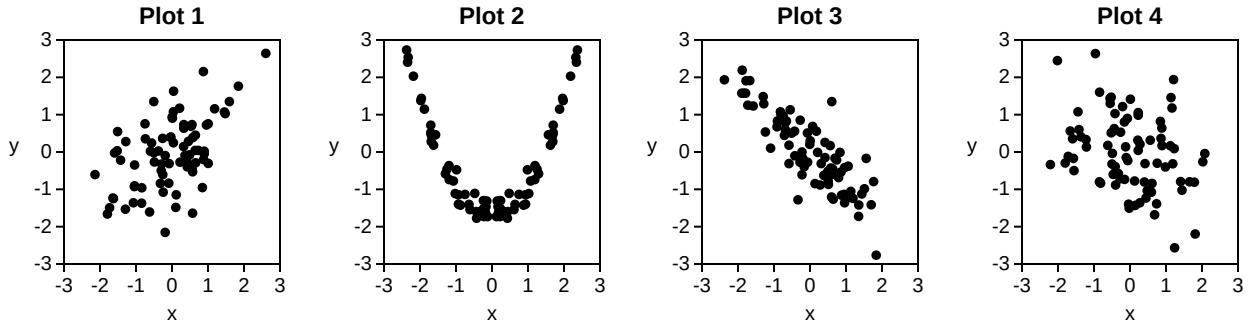
x	1	2	3	4	5
y	4	6	5	10	10

b) (5 points) Every x in part a) is replaced by $2x + 7$ and every y by $20 - 3y$. Find the new correlation and the new covariance.

c) (7 points) A sixth point $(6, k)$ is added to the five points in part a). Find k so that the correlation of all six points is exactly 0.

d) (5 points) A study of the 50 US states finds a correlation of 0.85 between a state's average years of schooling and its average income. A reporter concludes that for individual adults, the correlation between years of schooling and income is about 0.85. Comment.

e) (5 points) Match the correlations -0.85, -0.3, 0, and 0.6 to the scatter plots below. (No justification needed.)



6. (40 points) In a large course, homework averages (x) have mean 80 and SD 10, and final exam scores (y) have mean 70 and SD 15. The correlation is 0.6 and the scatter plot is football-shaped.

a) (6 points) Find the equation of the regression line for predicting the final exam score from the homework average.

b) (10 points) Among students with a homework average of 90, estimate the average final exam score, and estimate the percentage who scored below the class average of 70 on the final.

c) (10 points) A student is at the 10th percentile of homework averages. Predict the student's final exam score, and estimate the chance that the student scored above the median on the final.

d) (8 points) A student scored 100 on the final. Predict the student's homework average. Why can't you simply solve the equation from part a) for x?

e) (6 points) The instructor notices that the students in the bottom 10% on homework did better on the final, on average and in standard units, than they did on homework, while the top 10% did worse. She concludes that the final was "easier for weak students and harder for strong ones." Comment.

A Normal Table

Height = height of the standard normal curve at z (percent per standard unit). Area = percentage of the area under the normal curve between −z and z.

z	Height	Area	z	Height	Area	z	Height	Area
0.00	39.89	0.00	**1.50**	12.95	86.64	**3.00**	0.443	99.730
0.05	39.84	3.99	**1.55**	12.00	87.89	**3.05**	0.381	99.771
0.10	39.70	7.97	**1.60**	11.09	89.04	**3.10**	0.327	99.806
0.15	39.45	11.92	**1.65**	10.23	90.11	**3.15**	0.279	99.837
0.20	39.10	15.85	**1.70**	9.40	91.09	**3.20**	0.238	99.863
0.25	38.67	19.74	**1.75**	8.63	91.99	**3.25**	0.203	99.885
0.30	38.14	23.58	**1.80**	7.90	92.81	**3.30**	0.172	99.903
0.35	37.52	27.37	**1.85**	7.21	93.57	**3.35**	0.146	99.919
0.40	36.83	31.08	**1.90**	6.56	94.26	**3.40**	0.123	99.933
0.45	36.05	34.73	**1.95**	5.96	94.88	**3.45**	0.104	99.944
0.50	35.21	38.29	**2.00**	5.40	95.45	**3.50**	0.087	99.953
0.55	34.29	41.77	**2.05**	4.88	95.96	**3.55**	0.073	99.961
0.60	33.32	45.15	**2.10**	4.40	96.43	**3.60**	0.061	99.968
0.65	32.30	48.43	**2.15**	3.96	96.84	**3.65**	0.051	99.974
0.70	31.23	51.61	**2.20**	3.55	97.22	**3.70**	0.042	99.978
0.75	30.11	54.67	**2.25**	3.17	97.56	**3.75**	0.035	99.982
0.80	28.97	57.63	**2.30**	2.83	97.86	**3.80**	0.029	99.986
0.85	27.80	60.47	**2.35**	2.52	98.12	**3.85**	0.024	99.988
0.90	26.61	63.19	**2.40**	2.24	98.36	**3.90**	0.020	99.990
0.95	25.41	65.79	**2.45**	1.98	98.57	**3.95**	0.016	99.992
1.00	24.20	68.27	**2.50**	1.75	98.76	**4.00**	0.013	99.9937
1.05	22.99	70.63	**2.55**	1.54	98.92	**4.05**	0.011	99.9949
1.10	21.79	72.87	**2.60**	1.36	99.07	**4.10**	0.009	99.9959
1.15	20.59	74.99	**2.65**	1.19	99.20	**4.15**	0.007	99.9967
1.20	19.42	76.99	**2.70**	1.04	99.31	**4.20**	0.006	99.9973
1.25	18.26	78.87	**2.75**	0.91	99.40	**4.25**	0.005	99.9979
1.30	17.14	80.64	**2.80**	0.79	99.49	**4.30**	0.004	99.9983
1.35	16.04	82.30	**2.85**	0.69	99.56	**4.35**	0.003	99.9986
1.40	14.97	83.85	**2.90**	0.60	99.63	**4.40**	0.002	99.9989
1.45	13.94	85.29	**2.95**	0.51	99.68	**4.45**	0.002	99.9991